Sentinel Ridge Security

People-Process-Technology (PPT) Framework Analysis

Written to

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Purpose of the Analysis

This analysis applies the **People–Process–Technology (PPT)** framework to evaluate how cybersecurity and IT decisions should be structured for Gulf Coast Family Dental. The goal is to ensure that security controls are **effective, sustainable, and aligned with real business operations**, rather than purely technical ideals.

As a small healthcare practice handling sensitive patient and financial data, the organization faces meaningful cyber risk while operating under **budgetary, staffing, and operational constraints**. The PPT framework is used to balance these realities and guide defensible decision-making.

People

Organizational Reality

Gulf Coast Family Dental operates with:

- A **non-technical executive owner** (Dr. Lawson) who is the ultimate risk decision-maker
- A **clinically focused workforce** whose primary responsibility is patient care, not IT
- **High system usage** across clinical and administrative roles
- **No internal IT staff**, relying entirely on an outsourced contractor

Staff roles vary widely in technical comfort, authority, and exposure to sensitive data. This creates uneven risk profiles across the organization.

Key People-Based Risks

- **Phishing susceptibility**, especially among billing and hygienist staff
- **Privilege creep**, particularly for trusted or senior employees
- **Shared workstation usage**, increasing credential exposure
- **Overreliance on individuals** (e.g., office manager) with broad access
- Limited ability for staff to independently recognize or respond to cyber incidents

Most realistic threats stem from **human error rather than malicious intent.**

People-Focused Security Objectives

- Align system access strictly with job responsibilities
- Reduce reliance on individual judgment for security decisions
- Make secure behavior the *path of least resistance*
- Ensure leadership understands risk without needing technical expertise

Security controls must **compensate for human limitations**, not assume perfect behavior.

Process

Current State

The organization does not maintain a formally documented cybersecurity program. Many operational workflows are:

- Informal
- Knowledge-based (stored “in people’s heads”)
- Dependent on long-tenured staff

While this is common in small practices, it increases operational and security risk.

Critical Security Processes Identified

To support the business model, the following processes are essential:

1. **Access Management**
    - Role-based access provisioning
    - Periodic access review
    - Controlled privilege escalation
2. **Onboarding and Offboarding**
    - Standardized account creation
    - Immediate deactivation upon departure
    - Reduced risk from part-time and rotating staff
3. **Incident Response**
    - Clear contractor-led response model
    - Defined escalation paths to leadership
    - Predefined thresholds for downtime vs continuity
4. **Billing and Financial Verification**
    - Secondary validation for payment or insurance changes
    - Formal handling of “urgent” external requests
5. **Backup and Recovery**
    - Regular backups
    - Documented restoration procedures
    - Acceptance of recovery-time tradeoffs

Process Design Philosophy

Processes must be:

- **Simple enough to be followed consistently**
- Documented for handoff to future contractors
- Designed to minimize disruption to clinical care
- Aligned with the reality that IT expertise is external

A process that cannot be followed during a busy clinical day is effectively nonexistent.

Technology

Technology Constraints

Technology decisions are constrained by:

- A **$20,000 total project budget**
- Outsourced management and monitoring
- Limited tolerance for downtime
- Mixed-age hardware environments

As a result, technology must be **maintainable, resilient, and defensible**, rather than cutting-edge.

Technology's Role in the PPT Model

Technology is used to:

- Enforce access decisions made at the people and process level
- Reduce reliance on manual controls
- Increase visibility into high-risk activities
- Support incident detection and recovery

Importantly, technology is **not** expected to eliminate risk — only to reduce it to an acceptable level.

Technology Priorities

High-value technology investments include:

- Centralized identity and authentication
- Role-based access enforcement
- Multi-factor authentication for sensitive roles
- Endpoint protection suitable for SMB healthcare
- Network segmentation aligned with clinical workflows
- Logging and monitoring scaled to staffing realities

Lower-priority or deferred technologies are explicitly accepted as residual risk due to cost or complexity.

Interdependence of PPT Components

The three components are **mutually reinforcing**:

- **People** execute daily operations and are the most common source of security failure
- **Processes** guide behavior and reduce reliance on individual judgment
- **Technology** enforces processes and limits the impact of mistakes

A failure in any one area undermines the others. For example:

- Strong technology without training leads to workarounds
- Good processes without enforcement lead to inconsistency
- Trained people without tooling are eventually overwhelmed

Security effectiveness depends on **balance**, not strength in a single area.

Business Outcome of Applying PPT

By applying the PPT framework, Gulf Coast Family Dental achieves:

- Reduced likelihood of disruptive incidents (e.g., ransomware, billing fraud)
- Clear ownership of risk and decision-making
- Predictable operations during both normal and adverse conditions
- A security posture appropriate to its size, budget, and threat landscape

Most importantly, security becomes a **business enabler**, not an obstacle to patient care.